

Aditya Mate, PhD

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[Link to full CV](#)

Short C.V.

Summary

Applied Scientist at **Microsoft** and **Harvard PhD** (computer science) with **8 years of AI/ML research** experience. Research focuses on LLMs, reasoning agents, evaluation, safety, routing and decision-making under uncertainty. Prior research experience at **Google Research** and **IBM Research**. Published at NeurIPS, ICML, AAAI, AAMAS, EMNLP etc. with experience translating research into deployed ML systems in product settings.

Education

- 2018–2023 **Harvard University**, Cambridge, MA
Ph.D. in Computer Science | Thesis: Actualizing Impact of AI and Optimization in Public Health.
- 2013–2018 **Indian Institute of Technology Bombay**, Mumbai
B.Tech & M.Tech in Electrical Engineering | Minor: Computer Science and Engineering.

Industry Experience

- July'23– **Microsoft**, Cambridge, MA.
Present APPLIED SCIENTIST 2 | Manager: Dr. Yingnong Dang
- Currently focus on developing probabilistic models and LLM reasoning agents within Azure's AI Model & Solution team to improve the quality and reliability of Azure services.
 - Selected at Microsoft's [AI Incubation Program](#) by the Office of CTO, to lead infusion of AI/ML research in product problems. Collaborate with four product organizations across Microsoft: Strategic Missions and Technologies, Microsoft 365, GenAI, and Microsoft Search, Assistant and Intelligence (MSAI).
 - Led & mentored research and developed ML/LLM solutions across LLM evaluation, fine-tuning and safety, cost-aware LLM routing, RAG/Graph RAG, and reasoning over time-series data, translating research into deployed ML systems in product settings.
- 2021–2022 **Google Research**, United States / India.
RESEARCH INTERN / STUDENT RESEARCHER | Managers: Philip Nelson, Dr. Aparna Taneja
- Led the development of a novel approach for evaluating algorithmic resource-allocation policies in randomized trials, enabling more accurate causal estimates; published at [ICML 2023](#).
 - Led the field study and deployment of a restless multi-armed bandit system for improving maternal and child health outcomes, translating ML research into a real-world intervention program; published at [AAAI 2022](#) and recognized with an [IAAI 2023 Innovative Application Award](#).
 - Contributed to and mentored parallel research efforts in **decision-focused learning** and **sequential decision-making under uncertainty** and mentored related research projects, resulting in publications at [AAMAS 2023](#), [AAAI 2023](#), and EAAMO 2022.
- June–Sept'21 **IBM Research**, Yorktown Heights, NY (remote).
RESEARCH INTERN | Manager: Dr. Kush R. Varshney
- Data-driven methods for planning and resource allocation for social-impact non-profits ([paper](#)).

Selected Academic Achievements

- 2023 IAAI 'Innovative Application' Award for deployed AI system in maternal healthcare.
- 2022 INFORMS "[Doing Good with Good OR](#)" Competition Runner-up prize.
- 2021 **Best Paper Award** at [NeurIPS'21 Workshop](#) on Machine Learning in Public Health (MLPH).
- 2020 **Best Lightning Paper Award** at [NeurIPS'20 Workshop](#) on Machine Learning in Public Health.
- 2020 **Best Paper Award** (theme) at [NeurIPS'20 Workshop](#) on Machine Learning for Health (ML4H).

Selected Publications

Generative AI / LLM / Modern AI Systems

- CAIS 2026 Emmanuel Boateng, et al. **Aditya Mate**, Ehi Nosakhare.
"SAPO: Secure Automated Prompt Optimization via Multi-Agent Collaboration", *ACM Conference on AI and Agentic Systems (CAIS) 2026*, [\[paper\]](#).
- EMNLP 2025 Anh Pham et al., **Aditya Mate**, Ehimwenma Nosakhare, Soundararajan Srinivasan
"How to Fine-Tune Safely on a Budget: Model Adaptation Using Minimal Resources", *Empirical Methods in Natural Language Processing (EMNLP) 2025 Industry Track*, [\[paper\]](#).
- MSJAR 2025 Tian Xia, **Aditya Mate***, Regina Ceballos*, et al.
"Building Gold-Standard Datasets for AI Quality Evaluation", *Microsoft Journal on Applied Research (MSJAR) volume 22*, *Equal contribution, [MSFT confidential].
- NeurIPS 2025 workshop Roshini Pulishetty* et al., **Aditya Mate**, Somya Chatterjee et al.
"One Head, Many Models: Cross-Attention Routing for Cost-Aware LLM Selection", *Workshop on Evaluating the Evolving LLM Lifecycle NeurIPS 2025*, *Equal contribution.
- MSJAR 2023 Newman Cheng et al., **Aditya Mate**, Jonathan Larson
"Graph Based LLM Agents for Complex Reasoning Tasks", *Microsoft Journal of Applied Research (MSJAR) Volume 20, (2023)*, [\[paper internal to Microsoft\]](#).

ML / Reinforcement Learning / Decision-Making (All First Author)

- ICML 2023 **Aditya Mate**, Bryan Wilder, Aparna Taneja and Milind Tambe
"Improved Policy Evaluation for Randomized Trials of Algorithmic Resource Allocation", *International Conference on Machine Learning (ICML) 2023*, [\[paper\]](#).
- Book Chapter 2023 **Aditya Mate**, Shresth Verma, Gargi Singh, Aparna Taneja and Milind Tambe
"Field Study in Deploying Restless Multi-Armed Bandits: Assisting Non-Profits in Improving Maternal and Child Health", [\[link\]](#).
- AAAI 2022 **Aditya Mate***, Lovish Madaan*, Aparna Taneja, et al.
"Field Study and Deployment of Restless Multi-Armed Bandits for Improving Maternal and Child Health Outcomes", *AAAI Conference on Artificial Intelligence 2022*, *Equal contribution, [\[paper\]](#),
• Also appeared at **NeurIPS 2021 workshop on Machine Learning in Public Health (MLPH)**.
Won **Best Paper Award**
- AAMAS 2022 **Aditya Mate**, Arpita Biswas, Christoph Siebenbrunner, Susobhan Ghosh and Milind Tambe.
"Efficient Algorithms for Finite Horizon and Streaming Restless Multi-Armed Bandit Problems", *International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2022*, [\[paper\]](#).
- AAMAS 2021 **Aditya Mate**, Andrew Perrault and Milind Tambe.
"Risk-Sensitive Interventions in Public Health: Planning with Restless Multi-Armed Bandits", *International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2021*, [\[paper\]](#).
- NeurIPS 2020 **Aditya Mate***, Jackson Killian*, Haifeng Xu, Andrew Perrault and Milind Tambe.
"Collapsing Bandits and Their Application to Public Health Interventions", *Advances in Neural and Information Processing Systems (NeurIPS) 2020* *Equal contribution, [\[paper\]](#).

Professional Service & Invited Talks

- **Area Chair / Senior PC** for ICLR 2027, AAAI 2027, ICML (2026, 2025), NeurIPS 2026, IJCAI 2026, AAMAS 2025. Reviewer for several top-tier AI research conferences in prior years.
- **Invited Talks:** Carnegie Mellon University (Guest Lecture, course: "*Machine Learning in Practice*"); New York AI Agents Summit; Microsoft Research New England, ML Ideas Seminar;

Press Coverage & Research Impact

- Research covered in top-tier news outlets and AI blogs including: [Business Insider](#), [Google AI Blog](#), [Medical Xpress](#), [Nature Research Asia newsletter](#), [Harvard SEAS newsletter](#), [Sakal Media](#)